

ECE 1387 - CAD for Digital Circuit Synthesis and Layout
Assignment #2 – Analytical Placement and Flow-Based Spreading with Heterogeneous Cells

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Assignment Date: October 10, 2025
Due Date: October 24, 2025, 11:59pm, by electronic submission.
Late Penalty: -2 marks per day late, with total marks available = 20

You are to write an implementation of a basic analytical placer (AP), with overlap removal (spreading). As described in class, you will formulate the placement problem mathematically as a system of linear equations to be solved. You will use an existing package (UMFPACK within SuiteSparse) to solve the linear system (see Piazza page). For spreading, you will implement the flow-based approach, as described in class, and in the Darav et al. paper, “Multi-Commodity Flow-Based Spreading in a Commercial Analytic Placer,” ACM FPGA 2019.

Your program should display its progress and results using the same graphics package as used in Assignment #1. Your graphics should show the placement results and the connectivity between blocks (rat’s nest of wires). Blocks (cells) should appear as *squares* in your placement and be labeled with block numbers (see below).

The netlist file input format has two sections. The two sections are separated from one another by a -1 appearing by itself on a line. The first section specifies the blocks (cells) to be placed and the connectivity between them. Each line has the following form:

blocknum blocktype netnum₁ netnum₂ netnum₃ ... netnum_n -1

where blocknum is a positive integer giving the number of the cell, and blocktype is either 0 or 1. You will implement two styles of AP: In the first style (*homogeneous*), cells of either type can be placed *anywhere* on the chip. In the second style (*heterogeneous*), types 0 and 1 have restricted placement locations. Type 0 cells can be placed in any placement column i , where $i \bmod 4 = 0, 1, 2$. Type 1 cells can be placed in any placement column i , where $i \bmod 4 = 3$.

The netnum_i are the numbers of the nets that are attached to that block. Every block that has the same netnum_i on its description line is attached. Note that each block may have a different number of nets attached to it. Each line is terminated by a -1.

Example input file:

```
1 0 2 3 4 -1
2 0 5 4 -1
3 0 5 6 2 -1
4 1 6 3 -1
-1
1 40 0
4 0 40
-1
```

In this example, block 1 is type 0 and is connected to nets 2, 3 and 4. Note that each net may be connected to more than two blocks (that is, there are multi-fanout nets). Also, note that net numbers are not related to block numbers.

As discussed in class, the AP formulation requires there to be a set of pre-placed (fixed) objects, normally I/Os. The second section of the netlist file specifies the placement of fixed objects. It has the following form:

blocknum x_position y_position

In the above example, block 1 is pre-placed at the position with $x = 40$, $y = 0$. The list of fixed objects is terminated by a -1 by itself on a line. There may be multiple fixed objects placed at the same position (assume that is OK).

You should run your placer on the Assignment #2 test circuits provided on the course web page.

What to do and what to hand in?

Your placer must run on the ECF network. Instructions for electronic submission of your placer will be posted.

Your report should include a short description of the flow of your program, the main routines and what they do, assuming that we have basic knowledge of analytical placement.

1. **Do for all circuits:** Formulate and solve the analytical placement problem assuming the *clique* net model¹. Do not do any overlap removal in this step. Your program should display the placement and rat's nest (wires between cells) using graphics. Hand in a plot of the placement results. Your program should also compute the half-perimeter bounding box wirelength (HPWL) of the placement. *For computing HPWL, assume that nets connect to the centre of each cell.* Likewise, assume that the x, y position of each cell corresponds to the cell's centre. Use different colours in your graphics for the two different cell types. In your report, include a table showing the HPWL for each placed test circuit.
2. **Do for circuits 2 and 3 only for both the homogeneous and heterogeneous styles of AP:** Assuming that all bins are unit size (1×1), implement the flow-based spreading algorithm as described in class, and in Darav's 2019 paper. Assume that cell movements are quantized to multiples of 1 unit (meaning, when a cell is moved between two bins, its relative position in the original bin is the same as its relative position in the new bin). The placement area for all circuits is from 0,0 to 40,40 and as such, there are $40 \times 40 = 1600$ bins.
 - i) Experiment with different approaches for initializing and increasing the maximum-allowable cell movement parameter, ψ , in the outer loop of the spreading algorithm. In your report, comment on the approaches investigated and their impact on the total cell displacement between the solved and spread placements for both styles of AP. By total cell displacement, I mean the *sum* of the displacements across all cells between their original solved placement and their spread placement, see equation below. Report the total cell displacement for the approaches attempted. Report HPWL for the spread placement for both circuits, for both AP styles. In your report, explain how you handled the placement restrictions arising from the two different cell types.

$$TotalCellDisplacement = \sum_{i \in moveableCells} |x_{i,solved} - x_{i,spread}| + |y_{i,solved} - y_{i,spread}|$$

- ii) Introduce artificial *fixed* blocks (*anchors*), at each cell's re-placed (spread) position for the spreading approach you investigated in i) that provided the best (smallest) total cell displacement. Introduce an artificial two-pin net from the anchor to the corresponding cell. Experiment with *two different weight choices* for the artificial nets: weak and strong. The strong weight choice, for example, should be large enough to cause a cell to be pulled sharply towards its anchor when the system is re-solved. Solve the formulation corresponding to each of the two weight choices. Compute and *report* the half-perimeter bounding-box wirelength (HPWL) of the placements, and display and plot the results (there should be 8 plots as there are 2 AP styles). Include the plots in your report. Remember that you should not include the artificial nets in your HPWL calculation! Comment on what happens to HPWL when the new fixed blocks are introduced and the impact of the two different weight strengths.

¹ In the clique model, a net with p pins is represented as a complete graph (clique) with $p(p-1)/2$ edges. Each edge in the complete graph has weight of $2/p$. For example, a net with 2 pins has 1 edge with edge weight = 1. A net with 4 pins has 6 edges with edge weight = $2/4$.